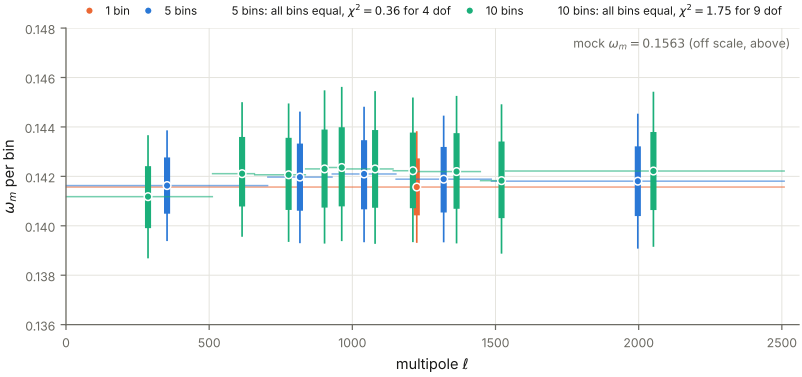
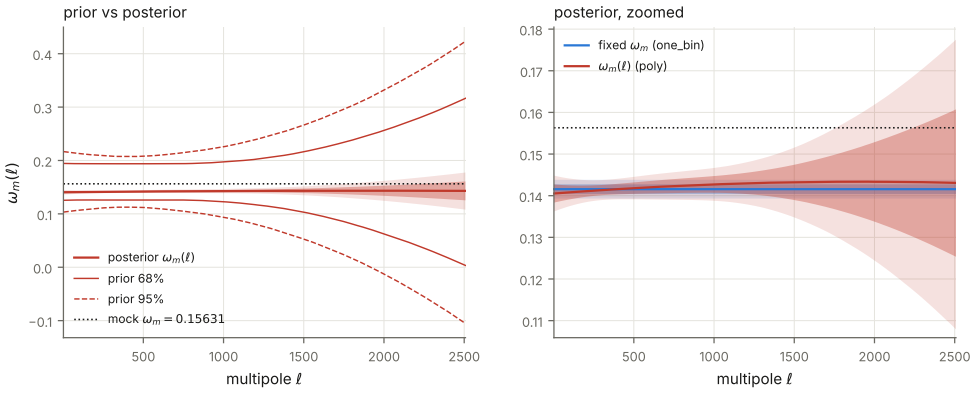
2-axion EDE mock: binned ω_m with 1, 5 and 10 bins



- Mock: mAxiCLASS 2-axion EDE ($n = 3$), Bella, Poulin, Vagnozzi, Knox 2026 PDH0 means: $f_{\rm EDE} = 0.085$, 0.073 at $\log_{10} a_c = -3.44$, -4.00; $H_0 = 72.6$, $\omega_m = 0.1563$. Fitted with `lcdm v1` (CosmoPower).
- All bins agree: $\omega_m \approx 0.1416$–0.1424 ($\sigma = 0.0012$–0.0016 per bin); test "all bins equal" $\chi^2 = 0.36$ (4 dof) and 1.75 (9 dof).
- ΛCDM absorbs the EDE into parameter shifts (1 bin, pulls vs mock): ω_m -12.7σ, h 0.679 (-8.4σ), n_s -5.5σ, ω_b -3.1σ; $\chi^2_{\rm min} = 5.35$.

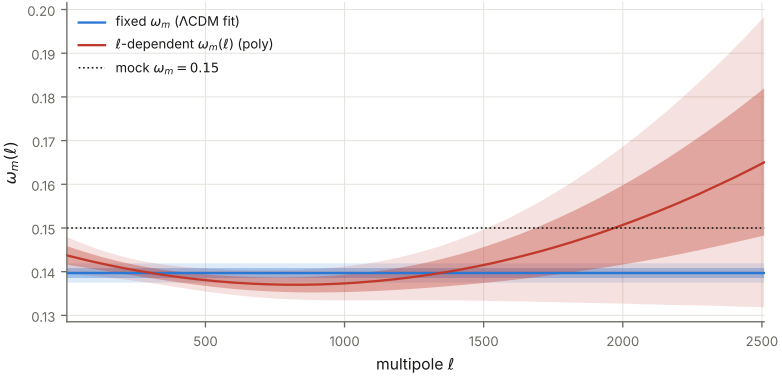
Runs of 2026-10-22 (after the mock fix). 8 chains $\times 10^5$ steps; acceptance 0.25 / 0.067 / 0.045, max $R - 1$ 0.0005 / 0.009 / 0.013. At 5–7% acceptance the chains' lowest χ^2 (5.83, 6.35) is not the true minimum.

2-axion EDE mock: smooth $\omega_m(\ell)$, prior vs posterior



- $\omega_{\boldsymbol{m}}(\ell) = \omega_{\boldsymbol{m0}} + b\,x + c\,x^2$, $x = (\ell - 400)/1000$; flat priors $\omega_{\boldsymbol{m0}} \in [0.11, 0.21]$, $b, c \in [-0.05, 0.05]$. The induced prior on $\omega_{\boldsymbol{m}}(\ell)$ is not flat and widens away from $\ell_{\boldsymbol{c}}$.
- Data-dominated everywhere: posterior 68% half-width 0.0013 ($\ell = 400$), 0.002 ($\ell = 1000$), 0.004 (1500), 0.018 (2500), vs prior 0.034, 0.037, 0.057, 0.156.
- No ℓ-dependence: $b = 0.0024 \pm 0.0040$, $c = -0.0008 \pm 0.0052$; $(b, c) = 0$ at $\chi^2 = 0.47$ (0.3σ); $\Delta\chi^2_{\rm min} = 0.40$ for 2 extra parameters.

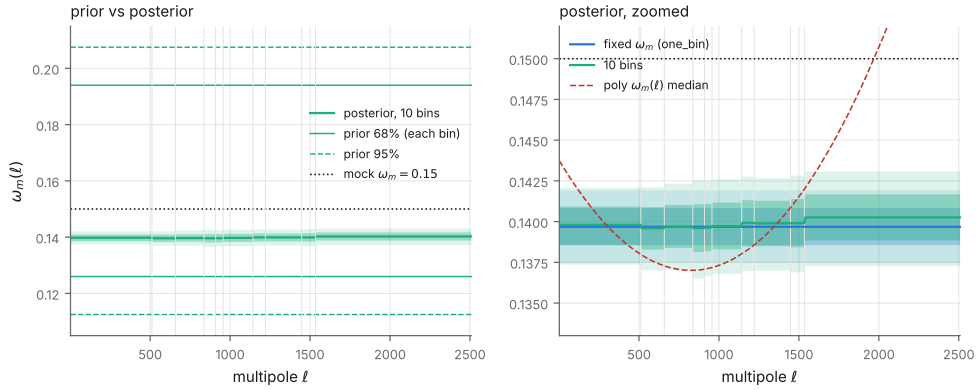
1-axion EDE mock (untuned): fixed ω_m vs $\omega_m(\ell)$



- Mock: the fiducial ΛCDM cosmology (ω_b 0.0224, h 0.674, τ 0.054, n_s 0.965, $\ln(10^{10}A_s)$ 3.044, ω_m 0.15) plus one axion ($n = 3$, $\theta_i = 2.8$), $f_{\rm EDE} = 0.12$ at $z_c = 10^4$. Nothing re-tuned: $100\theta_s = 1.031$ vs 1.049.
- ΛCDM fit: $\omega_m = 0.1397 \pm 0.0011$ (-9.1σ), $h = 0.641$ (-6.1σ), $n_s = 0.943$ (-6.2σ), ω_b -6.4σ; $\chi^2_{\rm min} = 13.8$.
- $\omega_m(\ell)$: $b = -0.0084 \pm 0.0039$, $c = 0.0099 \pm 0.0049$; $(b, c) = 0$ at $\chi^2 = 5.3$ for 2 dof (PTE 0.07, 1.8σ); $\Delta\chi^2_{\rm min} = 5.2$. Dip around $\ell \sim 800$, rise above $\ell \sim 1500$.

Runs of 2026-10-23. poly: 5 of 8 chains at 90–96k steps when copied (still running); max $R - 1$ 0.0009 (poly), 0.0004 (fixed).

1-axion EDE mock: 10 bins, prior vs posterior



- Flat prior $[0.11, 0.21]$ on each bin (68%: $[0.126, 0.194]$); posterior $\sigma = 0.0012$–0.0015 per bin: data-dominated in every bin.
- No deviation between bins: $\omega_m = 0.1396$–0.1402, test "all bins equal" $\chi^2 = 0.65$ for 9 dof. The bins follow the ΛCDM fit, not the curvature seen by the smooth model (right panel, dashed).

Preliminary: chains still running (89–100k steps when copied), acceptance 0.04, max $R - 1 = 0.043$; lowest χ^2 in the chains 15.8 (above the 1-bin 13.8: minimum not reached at this acceptance).